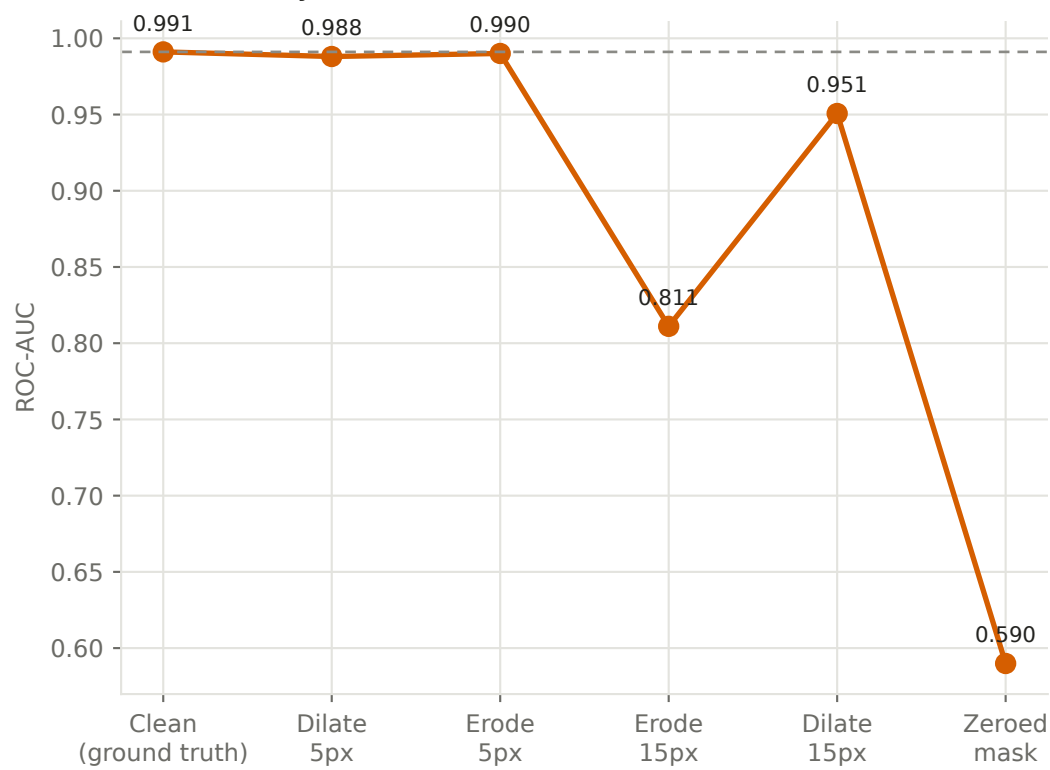
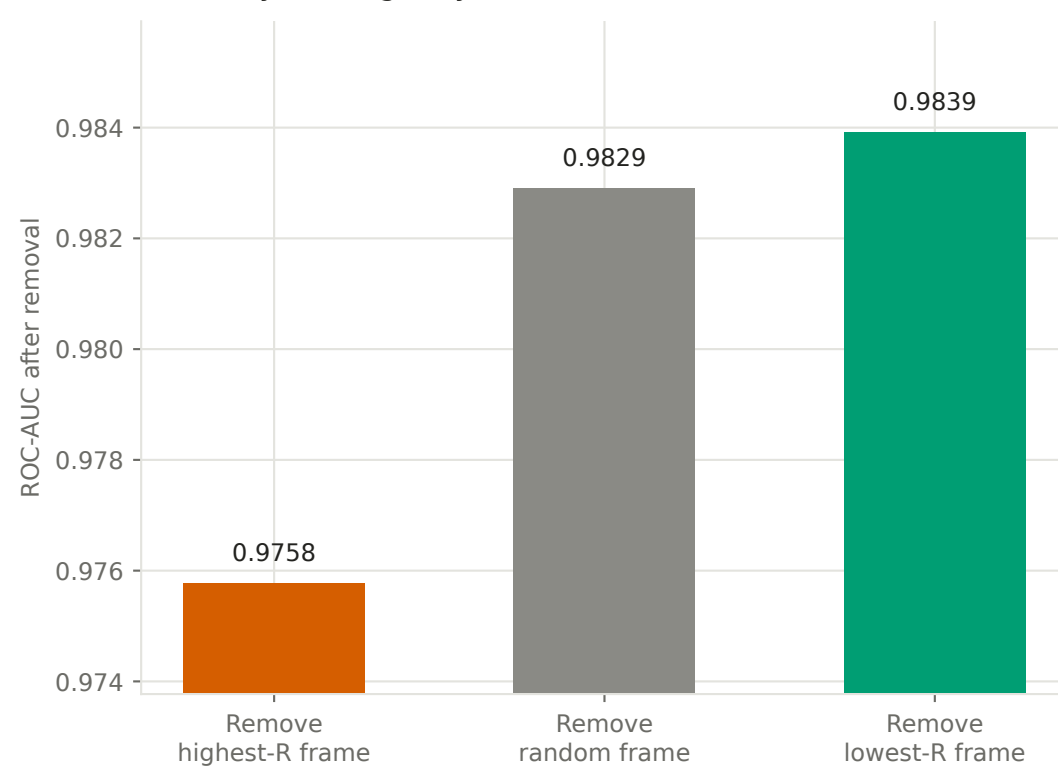


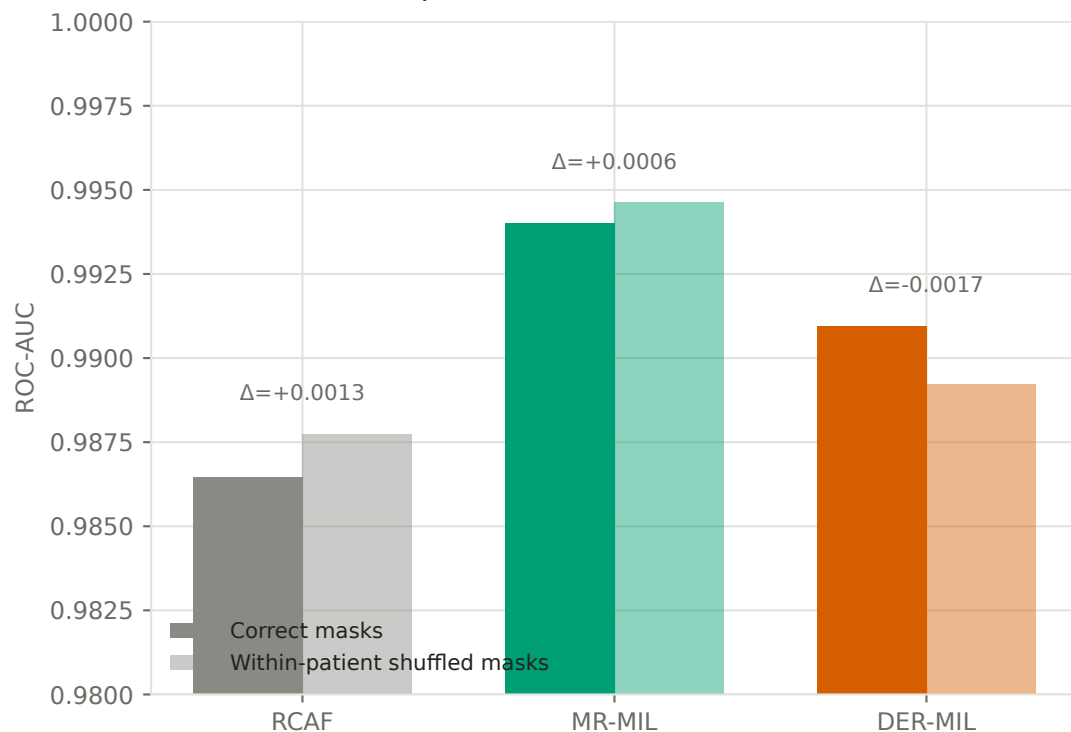
(a) Segmentation-quality degradation, no retraining
DER-MIL, ThyroidXL test



(b) One frame removed, by DER-MIL's own reliability ranking (ThyroidXL test)



(c) Within-patient mask-identity shuffle
(shortcut-reliance probe)



(d) Frame-order permutation invariance (DER-MIL)

Patients tested	707
Mean $ \Delta $ predicted probability	0.00e+00
95th-percentile $ \Delta p $	0.00e+00
Max $ \Delta p $ across all patients	0.00e+00

Shuffling the order in which a patient's frames are fed to the model changes its prediction by less than $1e-6$: the aggregation stage is exactly permutation invariant by construction, not just empirically stable.